

ALCOHOL'S, PHENOL'S AND ETHERS

ALCOHOL

- ♠ Alcohols are hydroxy derivatives of aliphatic hydrocarbons.
Eg: CH_3OH
- ♠ The functional group present in Alcohol and Phenol is the -OH (hydroxyl) group and that present in ethers is the -O- group (oxy group).
- ♠ In alcohols, the -OH group is bonded to an alkyl group. So, the general formula of alcohol is R-OH.

Classifications of Alcohols:-

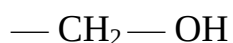
i) On the basis of the number of hydroxyl groups

- ♠ Based on this, alcohols are classified as monohydric (contain only one OH- group), dihydric (contain only two -OH group), trihydric (contain three -OH groups) and polyhydric (contain more than two -OH groups).
- ♠ Examples of monohydric alcohols are methanol (CH_3OH), ethanol ($\text{CH}_3\text{-CH}_2\text{-OH}$) etc.
- ♠ Example of dihydric alcohol is ethylene glycol ($\text{HO-CH}_2\text{-OH}$) and trihydric alcohol is glycerol ($\text{HOCH}_2\text{-CHOH-CH}_2\text{-OH}$)

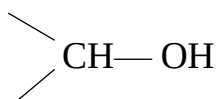
ii) According to the hybridisation of the C -atom, sp^3 or sp^2 to which the -The OH group is attached

1) Compounds containing sp^3 C-OH bond

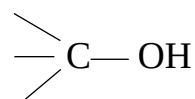
- ♠ Here the -OH group is attached to an sp^3 hybridised C-atom of the alkyl group.
- ♠ They are further classified as follows: Primary, secondary and tertiary alcohols.



Primary (1°)



Secondary (2°)

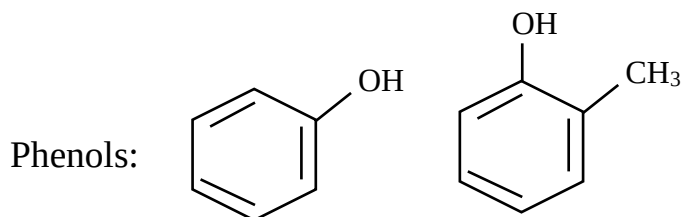


Tertiary (3°)

2) Compound containing sp^2 C-OH bond

- ♠ These alcohols contain -OH group bonded C—C double bond.

- ♠ These may be vinylic alcohols or phenols.



NOMENCLATURE

Common Name: By adding 'alcohol' to the name of the alkyl group (ie; word root + yl alcohol)

IUPAC Name: By replacing 'e' of alkane by 'ol' (ie; alkanol)

PREPARATION OF ALCOHOLS

1) From Alkenes

i) By acid-catalysed hydration

- ♠ Alkenes react with water in the presence of acids as catalysts to form alcohols.
- ♠ In the case of unsymmetrical alkenes, the addition reaction takes place in accordance with Markovnikov's rule.

Mechanism of this Reaction:- The mechanism of the reaction involves the following three steps:

Step 1: The H^+ ion present in water combines with water to form a hydronium ion (H_3O^+). The H_3O^+ ion then protonates the alkene to form a carbocation.

Step 2: Nucleophilic attack of water on carbocation.

Step 3: Deprotonation to form an alcohol.

ii) By hydroboration-oxidation reaction

- ♠ Alkenes add diborane to give trialkyl boranes as an addition product.
- ♠ This is on oxidation by hydrogen peroxide in the presence of aqueous sodium hydroxide to form alcohols.
- ♠ The net reaction is the addition of a water molecule to the alkene in a way opposite to Markovnikov's rule.

2) From Carbonyl Compounds

i) By the reduction of $RCHO$ & $RCOR$:

- ♠ Carbonyl compounds (aldehydes and ketones) when reduced using lithium aluminium hydride ($LiAlH_4$) or Sodium borohydride ($NaBH_4$) or on catalytic hydrogenation (using finely divided metal such as platinum, palladium or nickel), we get alcohols.
- ♠ Aldehydes give primary alcohols and ketones give secondary alcohols.

- ♠ Commercially, acids are reduced to alcohols by converting them to esters followed by their reduction using hydrogen in the presence of a catalyst (catalytic hydrogenation).

ii) By the reduction of carboxylic acid & ester

- ♠ esters can be reduced to primary alcohol by using $LiAlH_4$ but not with $NaBH_4$
- ♠ As $LiAlH_4$ is an expensive reagent commercially acids are reduced to alcohols by converting them to esters followed by their reduction($H_2/Ni/Pt/Pd$).

iii) From Grignard Reagents:

- ♠ Aldehydes and ketones add Grignard reagent followed by hydrolysis, we get alcohols.
- ♠ Formaldehyde (methanal) gives primary alcohols, aldehydes other than formaldehyde gives secondary alcohols and ketones gives tertiary alcohols.

3) From aliphatic 1° amines

PROPERTIES

Physical:

- ♣ **Boiling Point:** Alcohols have higher boiling than hydrocarbons, ethers and haloalkanes of comparable molecular masses.
- ♣ **Reason:** Due to the presence of intermolecular H-bonding in them.
- ♣ **Solubility:** Soluble in water.
- ♣ **Reason:** The ability to form intermolecular H-bonding with water but higher alcohols are H₂O insoluble as the hydrophobic hydrocarbon part predominates and resists H-bond formation with H₂O.

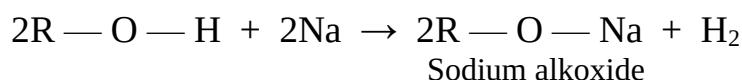
Chemical:

- ♣ Alcohols are acidic in nature.
- ♣ **Reason:** Due to the polar nature of the O-H bond.
- ♣ A releasing group (like -CH₃, -C₂H₅ etc.) increases the electron density of oxygen and hence decreases the polarity of the O-H bond. This decreases the acidic strength in the order: 1° alcohols > 2° alcohols > 3° alcohols

REACTIONS

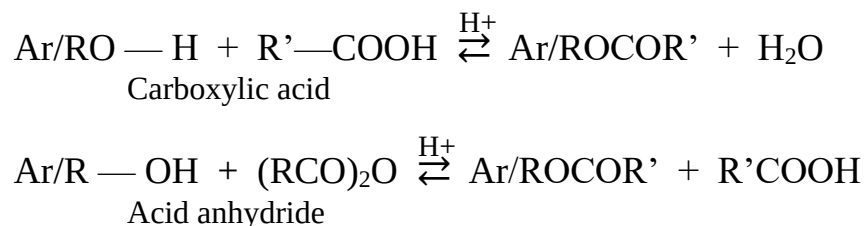
i) Reaction with Metals:

- ♠ Alcohols react with active metals such as sodium, potassium and aluminium to yield corresponding alkoxides.

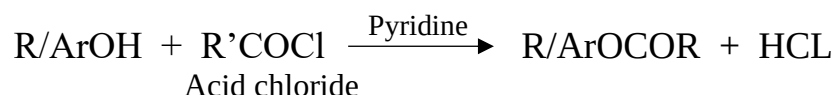


ii) Reaction with carboxylic acids, acid chlorides and anhydrides to form esters:

- ♠ The reaction with carboxylic acid and anhydride is carried out in the presence of conc. H_2SO_4 .



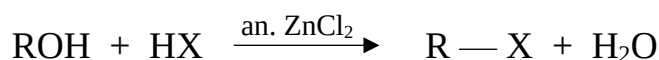
- ♠ The reaction with acid chloride is carried out in the presence of a base (pyridine) to neutralize the HCl formed during the reaction.



iii) Reaction involving cleavage of Carbon-oxygen (C-O) bond in alcohols

1) Reaction with hydrogen halides:

- ♠ Alcohols react with hydrogen halides in the presence of anhydrous zinc chloride (ZnCl_2) to form alkyl halides



- ♠ The reactivity of primary, secondary and tertiary alcohols with HCl is different; hence, this reaction is used to distinguish the three types of alcohols by a test called Lucas Test. The difference in reactivity of three classes of alcohols with HCl distinguishes them from one another (Lucas test).

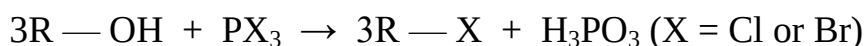
Lucas Test:

- ♣ Lucas reagent is a mixture of Conc. HCl and anhydrous ZnCl_2 .
- ♣ Alcohols are soluble in Lucas reagent while their halides are immiscible and produce turbidity in solution.

- ♣ Tertiary alcohols react with Lucas reagent and form immediate turbidity, secondary alcohols are less reactive and form turbidity within 5 minutes while primary alcohols do not produce turbidity at room temperature but they give turbidity on heating.

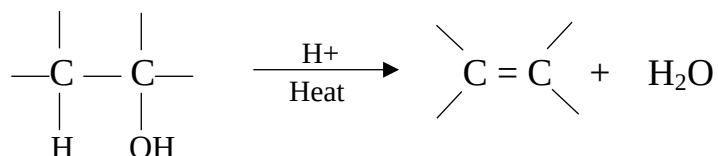
2) Reaction with phosphorous trihalides (PX₃):

Alcohols react with phosphorus trihalides to give alkyl halides.

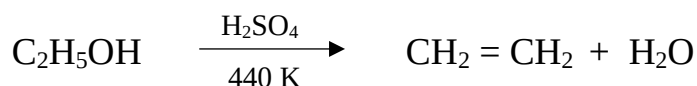


3) Dehydration:

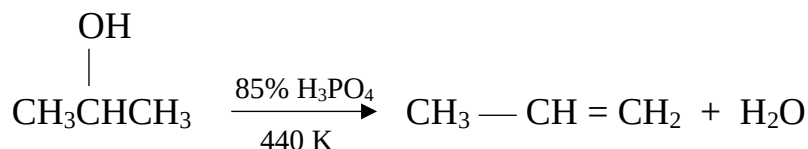
- ♠ Alcohols undergo dehydration (removal of a molecule of water) on treatment with a protic acid like concentrated H₂SO₄ or H₃PO₄, or catalysts such as anhydrous zinc chloride or alumina to form alkenes.

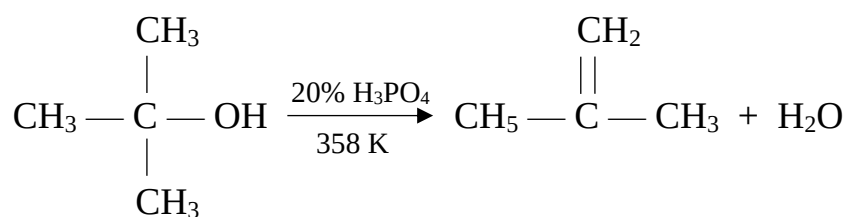


- ♠ The relative ease of dehydration of alcohols follows the order: Tertiary > Secondary > Primary. For example, ethanol undergoes dehydration by heating it with concentrated H₂SO₄ at 443 K, we get ethene.



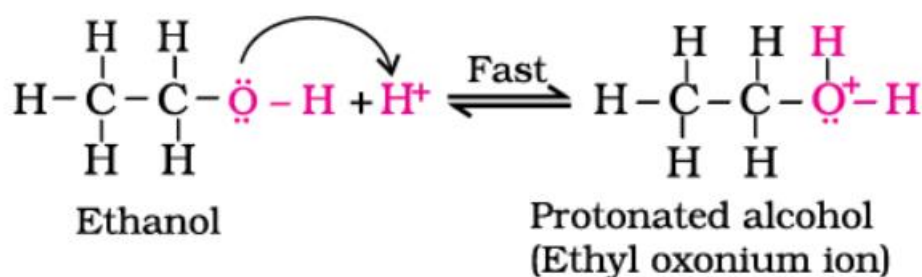
- ♠ Secondary and tertiary alcohols are dehydrated under milder conditions.





Mechanism of the reaction:- The reaction takes place in three steps.

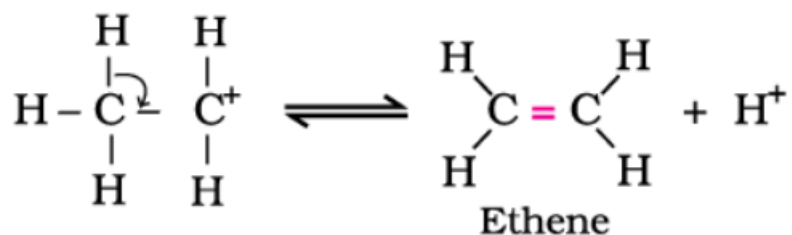
- ♣ **Step 1:** The alcohol on protonation gives an oxonium ion. This step is reversible and fast.



- ♣ **Step 2:** The oxonium ion loses a water molecule and forms a carbocation. This is the slowest step and hence, the rate-determining step of the reaction.



- ♣ **Step 3:** Formation of ethene by elimination of a proton.



The acid used in step 1 is released in step 3.

4) Oxidation:

- ♠ Alcohols on oxidation give carbonyl compounds (aldehydes and ketones) or carboxylic acids depending on the nature of the oxidising agent used.
- ♠ Primary alcohols when oxidized using mild oxidising agents like anhydrous CrO_3 we get aldehydes. But with strong oxidising agents like acidified potassium permanganate or potassium dichromate, carboxylic acids are formed



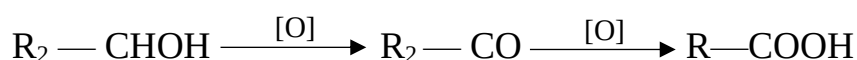
- ♠ Primary alcohols are oxidised to aldehydes in good yield by pyridinium chlorochromate (PCC) [a complex of chromium trioxide with pyridine and HCl].



- ♠ Secondary alcohols are oxidised to ketones by chromic anhydride (CrO_3).



- ♠ With strong oxidising agents, secondary alcohols give carboxylic acids with a lesser number of carbon atoms.

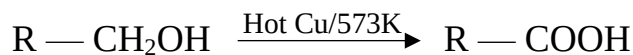


- ♠ Tertiary alcohols do not readily undergo oxidation reactions. But in the presence of strong oxidising agents (KMnO_4) and at high temperatures, they first give ketones with a lesser number of carbon atoms which on further oxidation give carboxylic acids with still lesser number of carbon atoms.

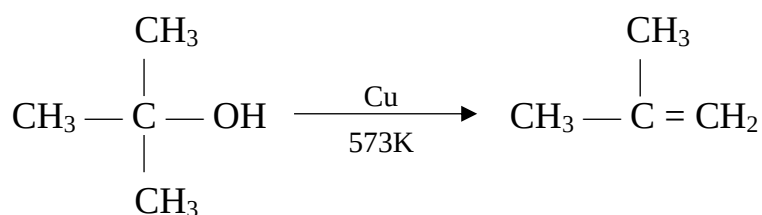


5) Reaction with hot copper catalyst:

- ♠ Primary alcoholic vapours when passed through hot Cu catalyst at 573K, undergo dehydrogenation to form aldehydes, while secondary alcohols undergo dehydrogenation to give ketones.



♠ Tertiary alcohols react with a hot Cu catalyst at 573 K and undergo dehydration to give alkenes.



IDENTIFICATION OF 1°, 2° AND 3° ALCOHOLS

	Test	1° alcohol	2° alcohol	3° alcohol
1	Reaction with Lucas reagent (HCl + ZnCl ₂)	No turbidity on heating. Give turbidity on heating.	Produce turbidity within 5 minutes.	Produce turbidity immediately
2	Oxidation with mild oxidising agents	Yield aldehydes	Yield ketones	No reaction (Resistant to oxidation)
3	Oxidation with 3 strong oxidising agents	Yield carboxylic acids	Yield ketones	No reaction. At elevated temp, yield carboxylic acids with a lesser number of C-atoms
4	Vapours when 4 passed through hot Cu at 573K	Yield aldehydes	Yield ketones	Yield alkenes

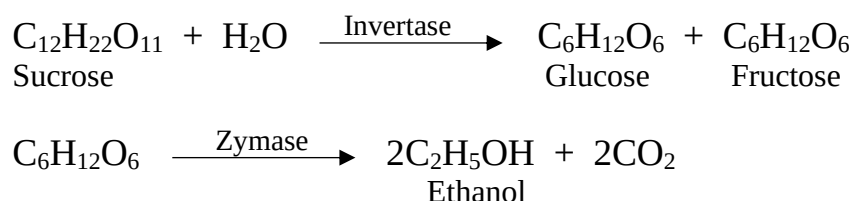
Some Commercially Important Alcohols

1) Methanol (CH₃OH)

- ♠ Methanol is also known as 'wood spirit' since it can be produced by the destructive distillation of wood. It is manufactured by the catalytic hydrogenation of carbon monoxide at about 573-673 K temperature and 200-300 atm pressure and in the presence of $\text{ZnO} - \text{Cr}_2\text{O}_3$ catalyst.
- ♠ It is highly poisonous in nature. It is used as a solvent in paints, varnishes and for making formaldehyde.

2) Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)

- ♠ Ethanol is commonly known as spirit or grain alcohol. It is obtained commercially by the fermentation of sugar. The sugar in molasses, sugarcane or fruits like grapes is converted to glucose and fructose, in the presence of an enzyme, invertase. Glucose and fructose undergo fermentation in the presence of another enzyme, zymase to give ethanol and carbon dioxide. Both the enzymes invertase and zymase are produced by yeast.



- ♠ Fermentation takes place in anaerobic conditions, i.e., in the absence of air. If air gets into the fermentation mixture, the oxygen of air oxidises ethanol to ethanoic acid (acetic acid), which destroys the taste of alcohol.
- ♠ The ethanol solution obtained by fermentation contains only 8-10% ethanol and it is called wash. It can be concentrated up to 95.6% by fractional distillation. 95.6% alcohol is called Rectified Spirit. It is concentrated to 100% by distilling with a small amount of benzene. Ethanol-free from water and other impurities (100% pure ethanol) is known as absolute alcohol.
- ♠ Ethanol is a colourless liquid. It is used as a solvent in the paint industry and the preparation of a number of carbon compounds.
- ♠ The commercial alcohol is made unfit for drinking by mixing in it some copper sulphate (to give it a colour) and pyridine (a foul-smelling liquid). It is known as denaturation of alcohol and the

resulting alcohol is known as denatured spirit. If methanol is used for denaturation, it is called methylated spirit.

♠ Ethanol can also be manufactured by hydration of ethene.



PHENOLS

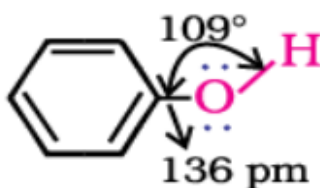
Phenols are hydroxy derivatives of aromatic hydrocarbons in which the -OH group(s) are directly attached to the C-atom of the ring.

Eg: $\text{C}_6\text{H}_5\text{OH}$

ELECTRONIC STRUCTURE OF HYDROXYL GROUP

In phenols, the -OH group is attached to sp^2 hybridised carbon of an aromatic ring. The C- O bond length (136 pm) in phenol is slightly less than that in alcohol. This is due to –

- (i) Partial double bond character of unshared electron pair of oxygen with the aromatic ring and
- (ii) sp^2 hybridised state of carbon to which oxygen is attached.



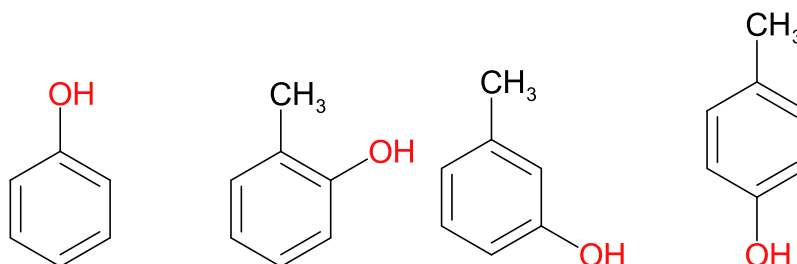
Phenol

NOMENCLATURE

Common Names: The parent member is Phenol. The position of the substituents is indicated by the prefixes ortho (for 1,2-disubstituted), meta (for 1, 3) and para (for 1, 4).

NB: Some substituted phenols are popularly known by their trivial names. Eg: Cresol (hydroxy derivatives of toluene)

IUPAC Name: Named as derivatives of phenols. The position of the substituents with respect to the -OH group is indicated by arabic numerals (with the carbon carrying -OH group being numbered 1).



Common Name: Phenol

o-cresol

m-cresol

p-cresol

IUPAC Name: Phenol

2-Methylphenol

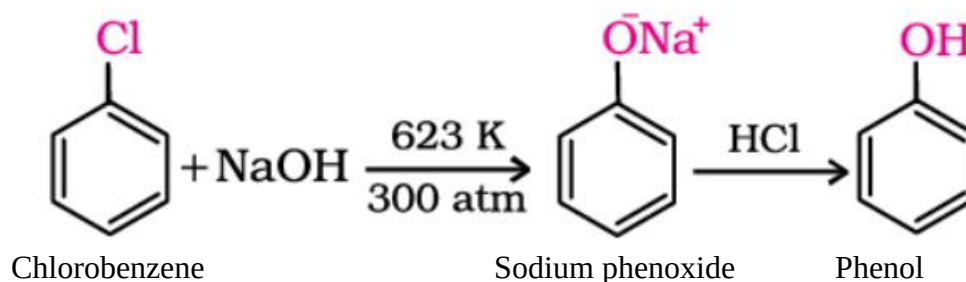
3-Methylphenol

4-Methylphenol

PREPARATION OF PHENOLS

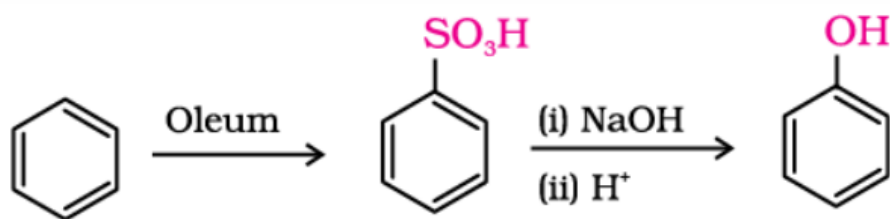
1) From haloarenes:

When chlorobenzene is fused with NaOH at 623K temperature and 320 atmospheric pressure, sodium phenoxide is formed which on acidification, we get phenol.



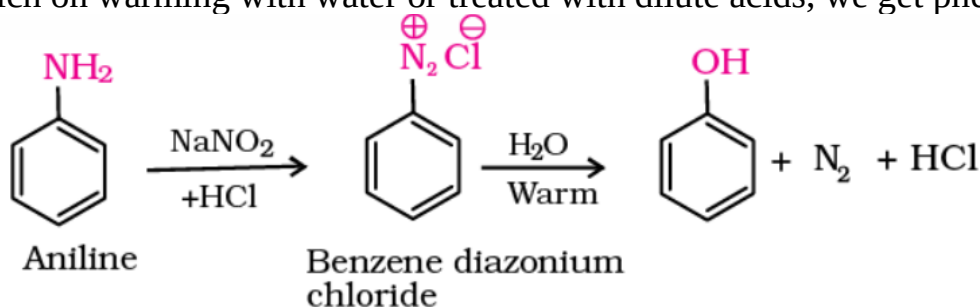
2) From benzene sulphonic acid:

Benzene on sulphonation with oleum, benzene sulphonic acid is formed which on treatment with NaOH followed by acidification, we get phenol.



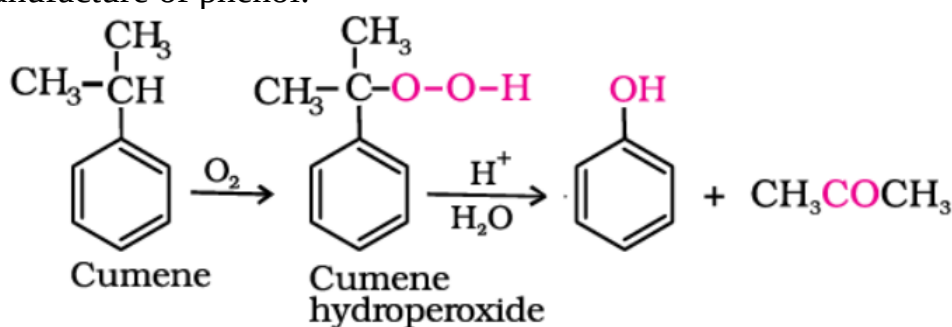
3) From diazonium salts:

When an aromatic primary amine (e.g. aniline) is treated with nitrous acid (prepared by mixing NaNO_2 & HCl) at 273-278K, a diazonium salt is formed, which on warming with water or treated with dilute acids, we get phenol.



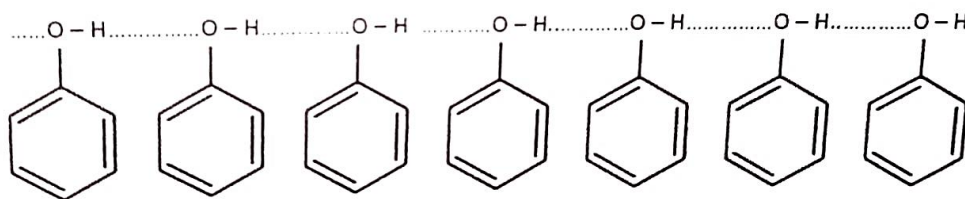
4) From cumene:

Cumene is isopropyl benzene (2-phenyl propane). It is oxidised in the presence of air, we get cumene hydroperoxide, which on treatment with dilute acid, phenol and acetone is formed. This method is used for the manufacture of phenol.



PHYSICAL PROPERTIES

- Boiling Points:** The boiling points of phenols are higher than hydrocarbons, ethers, haloalkanes and haloarenes of comparable molecular masses. This is because the -OH group phenols are involved in intermolecular hydrogen bonding.

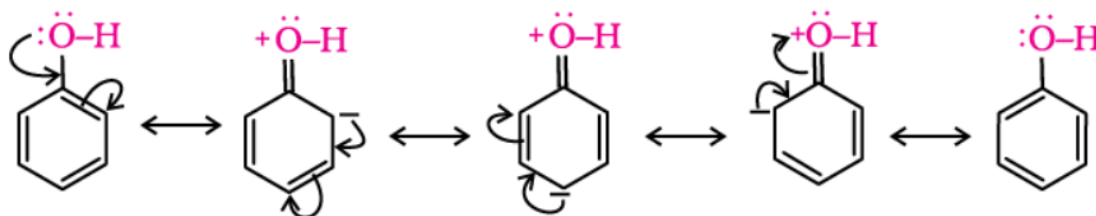


The boiling points of phenols increase with the increase in the number of carbon atoms. This is because as the number of carbon atoms increases, the van der Waals forces of attraction increase and hence the boiling point.

- 2) **Solubility:** Phenols are soluble in water. The solubility of alcohols and phenols in water is due to their ability to form hydrogen bonds with water molecules (inter-molecular hydrogen bonding).

CHEMICAL PROPERTIES

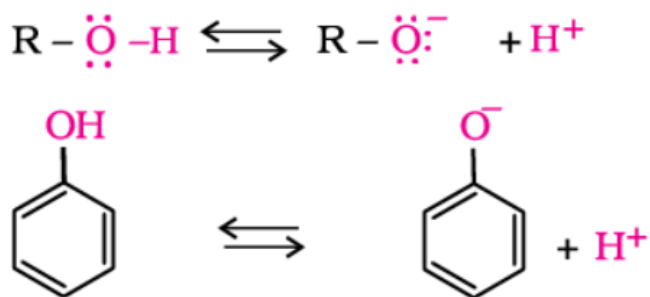
Acidity of Phenols: In phenols, the -OH group is directly bonded to an sp^2 hybridized carbon atom of the benzene ring. Due to the greater electronegativity of sp^2 hybridized carbon (due to greater s-character). The benzene ring acts as an electron-withdrawing group. So the lone pair electrons present in the oxygen atom of the -OH group enter the benzene ring and the following resonating structures are obtained:



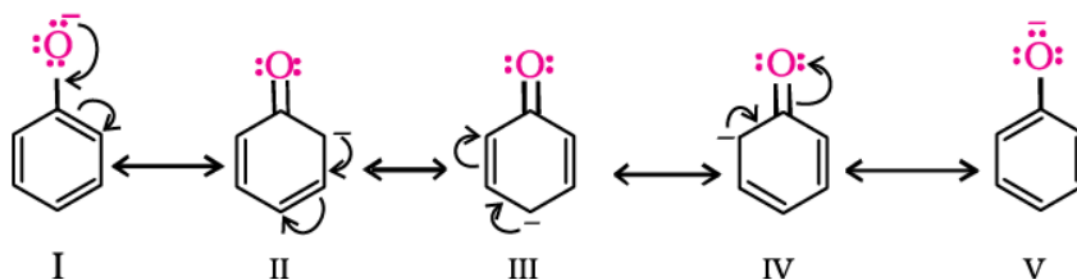
In the resonating structures II, III and IV, there is a positive charge on the electronegative oxygen atom and hence it attracts the bond pair of electrons in the O-H bond. So, it is easy to remove the hydrogen atom as an H^+ ion and thus phenol is acidic in nature.

Phenol is more acidic than alcohol. This is due to the following reasons:

1. In alcohol, the O-H group is directly bonded to an sp^3 hybridized carbon atom, but in phenol, it is bonded to an sp^2 hybridized carbon. Due to the greater s-character and electronegativity of sp^2 hybridized carbon, the ease of O-H bond cleavage is greater on phenol and hence it is more acidic than alcohol.
2. The ionization of alcohol and phenol is as follows:



The alkoxide ion ($R-O^-$) formed by the ionization of alcohol is not resonance stabilized. So the negative charge is localized on the oxygen atom. But the phenoxide ion ($C_6H_5O^-$) formed by the ionization of phenol is resonance stabilized as follows:



The negative charge is delocalized and hence phenoxide ion is more stable which favours the ionization of phenol. Also, phenoxide ion is more stable than phenol, because in phenol, there is a +ve charge on the electronegative oxygen atom. So, it is less stable and readily loses H^+ ion.

The presence of electron-withdrawing groups (like the nitro group) at ortho and para positions increases the acidic strength of phenol. It is due to the effective delocalisation of negative charge in phenoxide ion. On the other hand, electron-releasing groups (like alkyl groups) at these positions decrease the acidic strength of phenol. So cresols are less acidic than phenol.

REACTIONS OF PHENOLS

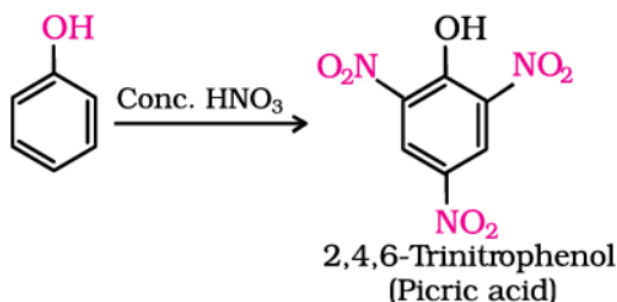
1) Electrophilic Substitution Reaction

In phenol, the -OH group attached to the benzene ring donates electron pairs and hence it activates it towards electrophilic substitution. Also, in the resonating structures of phenol, the electron density is greater in ortho and para positions. So the electrophile enters at these positions.

The common electrophilic aromatic substitution reactions taking place in phenol are:

a) Nitration:

- ◆ Phenol reacts with Conc. Nitric acid gives a yellow precipitate of 2,4,6-trinitrophenol commonly called Picric acid.



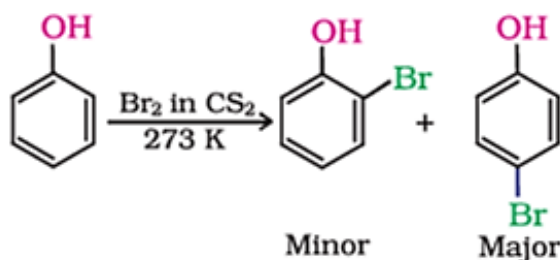
- ◆ Picric acid can also be prepared by treating phenol first with concentrated sulphuric acid followed by treating it with concentrated nitric acid.
- ◆ For the preparation of ortho and para nitrophenols, phenol is treated with dil. HNO_3 at low temperature (298K).



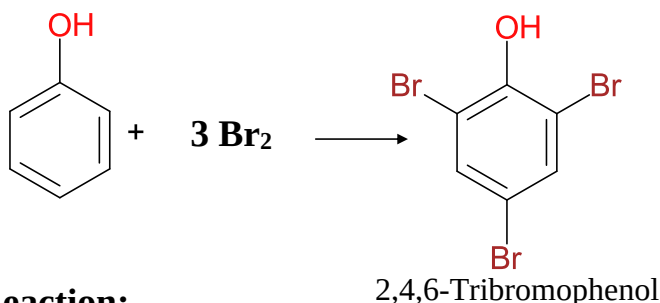
- ◆ The ortho and para isomers can be separated by steam distillation, o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding while p-nitrophenol is less volatile due to intermolecular hydrogen bonding which causes the association of molecules.

b) Halogenation:

- ◆ When phenol is brominated using Br_2 in CHCl_3 or CS_2 at low temperature, we get a mixture of ortho and para bromophenols. The reaction takes place in the absence of Lewis acid catalyst (like FeBr_3). It is due to the highly activating effect of the $-\text{OH}$ group attached to the benzene ring.

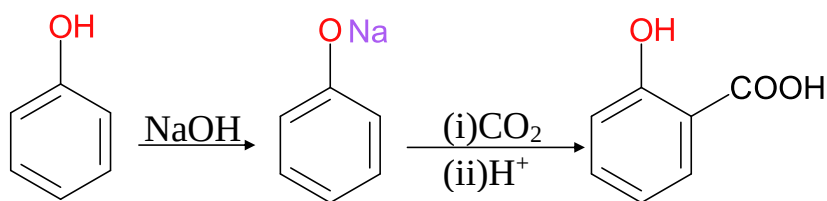


- ◆ When phenol is treated with bromine water, we get a white precipitate of 2,4,6-tribromophenol.



c) Kolbe's Reaction:

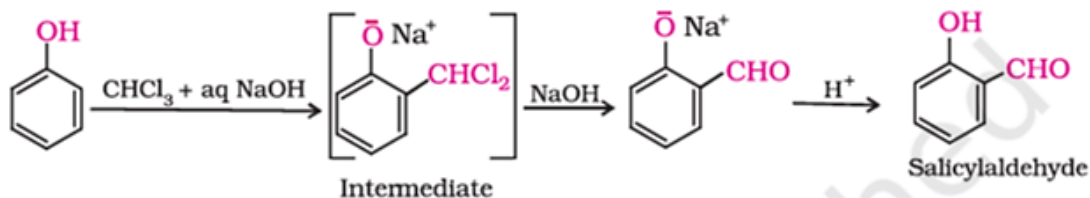
- ◆ Phenol when treated with sodium hydroxide, we get sodium phenoxide which on treatment with carbon dioxide followed by acidification, we get orthohydroxybenzoic acid commonly called Salicylic acid. This reaction is called Kolbe's reaction.



2-Hydroxybenzoic acid (Salicylic acid)

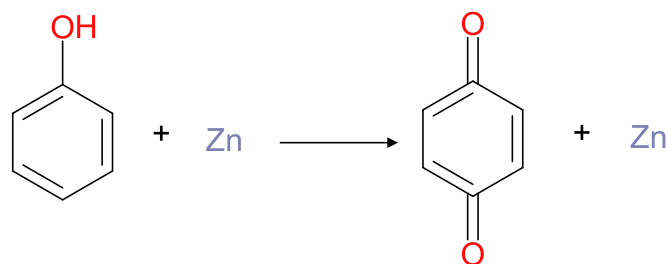
d) Reimer-Tiemann Reaction:

- ◆ Phenol when treated with chloroform in the presence of sodium hydroxide, followed by acidification, we get salicylaldehyde (o-hydroxybenzaldehyde). This reaction is known as Reimer - Tiemann reaction.



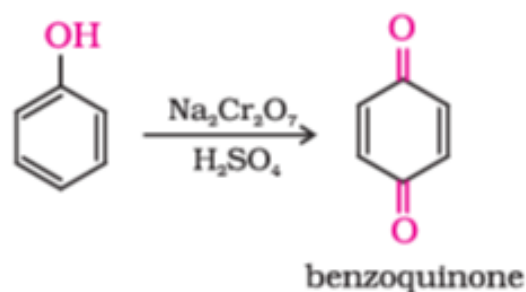
e) Reaction with zinc dust:

- ◆ Phenol when heated with Zn dust, we get benzene.



f) Oxidation:

- ◆ When phenol is oxidised with chromic acid, we get a conjugated diketone known as benzoquinone.
- ◆ In the presence of air, phenols are slowly oxidised to dark-coloured mixtures containing quinones.



Distinction between Alcohols and Phenols

- ♣ **Litmus Test:** Phenols turn blue litmus red but alcohols do not.
- ♣ **Bromine water Test:** Phenol on shaking with bromine water gives a white ppt. of 2,4,6-tribromophenol but alcohols do not

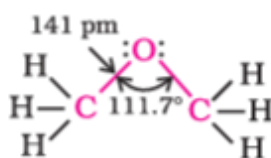
ETHERS

Ethers are compounds in which the H-atoms of a hydrocarbon are substituted by alkoxy (R-O) or aryloxy (Ar-O) groups.

Eg. CH_3OCH_3

ELECTRONIC STRUCTURE

- ♣ In ethers, the 4 electron pairs, (the 2 bond pairs and 2 lone pairs) on oxygen are arranged approximately in a tetrahedral arrangement.
- ♣ The bond angle is slightly greater than the tetrahedral angle due to the repulsive interaction between the two bulky (-R) groups.



Methoxymethane

CLASSIFICATION

- ♣ Ethers are classified on the basis of groups attached to the O-atom.
- ♣ Simple/symmetrical- The alkyl or aryl groups attached to the O-atom are the same.
Eg: Diethyl ether $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
- ♣ Mixed/unsymmetrical- The alkyl or aryl groups attached to the O-atom are different.
Eg: $\text{C}_2\text{H}_5\text{OCH}_3$ or $\text{C}_2\text{H}_5\text{OC}_6\text{H}_5$

NOMENCLATURE

- ♣ **Common Name:** By adding the word 'ether' to the names of alkyl/aryl groups written as separate words in alphabetical order.
Eg: Ethylmethyl ether: $\text{CH}_3\text{-O-C}_2\text{H}_5$
- ♣ If both the alkyl groups are the same, the prefix 'di' is added before the alkyl group.
Eg: Diethyl ether: $\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5$
- ♣ **IUPAC Name:** As 'Alkoxyalkane'. The larger alkyl (R) group is chosen as the parent hydrocarbon.

Compound	Common Name	IUPAC Name
CH_3OCH_3	Dimethyl ether	Methoxymethane
$\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	Diethyl ether	Ethoxyethane

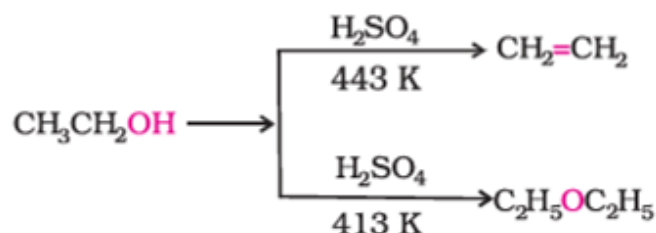
$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$	Methyl n-propyl ether	1-Methoxypropane
$\text{C}_6\text{H}_5\text{OCH}_3$	Methylphenyl ether (Anisole)	Methoxybenzene (Anisole)
$\text{C}_6\text{H}_5\text{OCH}_2\text{CH}_3$	Ethylphenyl ether (Phenetole)	Ethoxybenzene
$\text{C}_6\text{H}_5\text{O}(\text{CH}_2)_6\text{—CH}_3$	Heptylphenyl ether	1-Phenoxyheptane

- ♣ If the two alkyl groups on both sides of the -O- group are identical, it is called symmetric ether and if they are different, it is called asymmetric ether.

PREPARATION OF ETHERS

1) By dehydration of Alcohols

- ♠ Alcohols undergo dehydration in the presence of protic acids like H_2SO_4 or H_3PO_4 to give ethers based on the reaction conditions.
For example, ethanol when dehydrated with Conc. H_2SO_4 at 413 K, we get diethyl ether but at 443 K, we get ethene.



Mechanism: The formation of ether follows $\text{S}_{\text{N}}2$ mechanisms and involves three steps:

- ♦ In the first step, one molecule of the alcohol adds a proton to get a protonated alcohol. In the second step, this protonated alcohol is attacked by another molecule of alcohol to get an oxonium ion. In the last step, the oxonium ion loses a proton to form the ether.

2) Williamson's Ether Synthesis

- ♠ Alkyl halide reacts with sodium alkoxide to form ether.



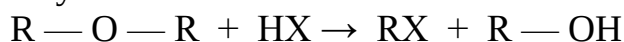
PHYSICAL PROPERTIES

- ♣ Ethers have a lower boiling point than alcohols. This is because in alcohols, there is intermolecular hydrogen bonding, which is absent in ethers.
- ♣ Lower ethers are miscible with water due to the formation of hydrogen bonding with water.

CHEMICAL REACTIONS

i) Reaction with HX

- ♠ Ethers in reaction with Hydrogen halide, cleave the C-O bond and form alkyl halides.



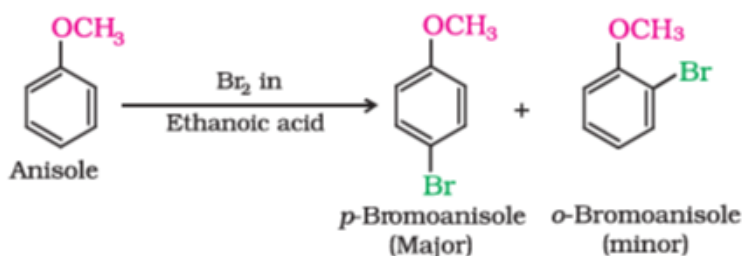
- ♠ In the case of ethers containing two different alkyl groups (Primary and secondary), the lower alkyl group forms the alkyl halide (1° alkyl halide)

ii) Electrophilic Substitution Reactions

- ♠ The alkoxy group (-OR) is ortho, para directing and activates the benzene ring. So, the product formed is a mixture of ortho and para isomers.

a) Halogenation:

Anisole on bromination with bromine in ethanoic acid to give a mixture of o-bromoanisole and p-bromoanisole.



b) Nitration:

Anisole reacts with a mixture of conc. sulphuric acid and conc. Nitric acid to give a mixture of ortho and para nitroanisole.



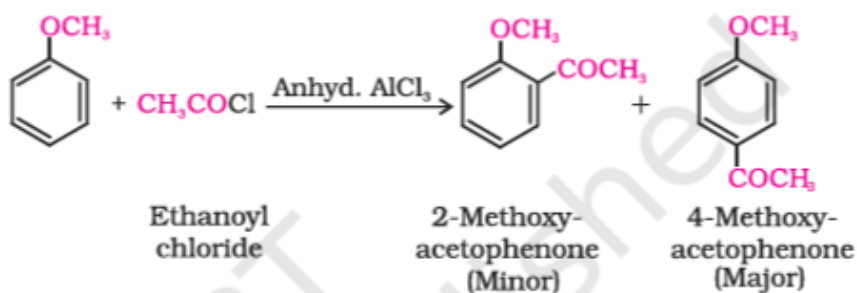
c) Friedel-Crafts Reaction:

i) Alkylation:

Anisole undergoes Friedel-Crafts alkylation reaction in the presence of anhydrous aluminium chloride (a Lewis acid) as a catalyst, we get a mixture of ortho and para-alkylated anisole.



ii) Acylation:



QUESTIONS & ANSWER

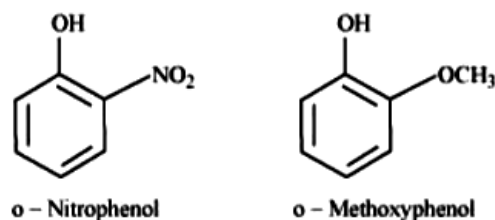
1) While separating a mixture of ortho and para nitro phenols by steam distillation, name the isomer which will be steam-volatile. Give reason.

Ans: O-nitrophenol is steam volatile due to intramolecular hydrogen bonding and hence can be separated by steam distillation from p-nitrophenol which is not steam volatile because of inter-molecular hydrogen bonding.

2) Explain why ortho-nitrophenol is more acidic than ortho-methoxyphenol.

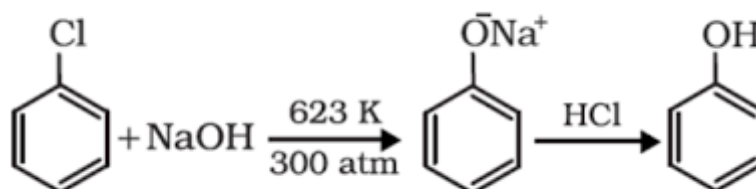
Ans: The nitro-group is an electron-withdrawing group. Therefore, decreases the electron density of the ring as well as oxygen. As a result, it is easier to lose a proton. Also, the o-nitrophenoxide ion formed after the loss of proton is stabilized by resonance. Hence, o-nitro phenol is a stronger acid. On the other hand, the methoxy group is an electron-releasing group. Thus, it increases the electron density of the

oxygen atom and hence, the proton cannot be given out easily. Therefore, o-nitrophenol is more acidic than o-methoxyphenol.



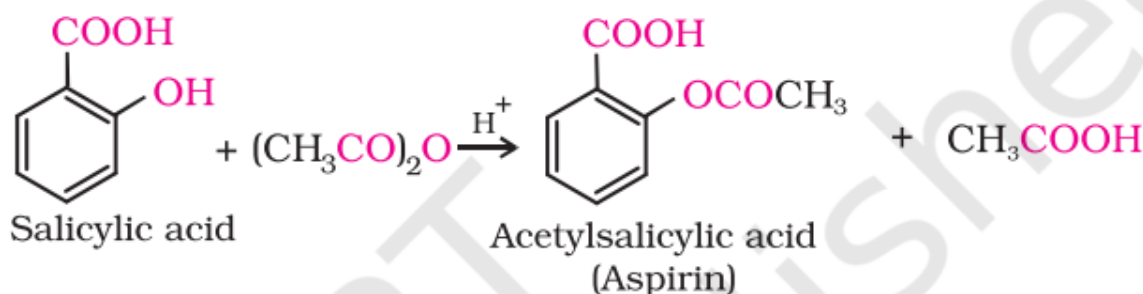
3) Write a chemical reaction for the preparation of phenol from chlorobenzene.

Ans: Chlorobenzene is fused with NaOH at 623K and 320 atmospheric pressure. Phenol is obtained by acidification of sodium phenoxide so produced.



4) How is aspirin (Acetylsalicylic acid) prepared from salicylic acid?

Ans: Acetylation of salicylic acid produces aspirin.



5) Which out of propan-1-ol and propan-2-ol is stronger acid?

Ans: Propan-1-ol is a stronger acid than propan-2-ol. The acidic strength of alcohols is in the order $1^\circ > 2^\circ > 3^\circ$.

6) What is the denaturation of an alcohol?

Ans: The commercial alcohol is made unfit for drinking by mixing it with some copper sulphate (to give it a colour) and pyridine (a foul smelling liquid). It is known as the denaturation of alcohol.

7) Give IUPAC name of $\text{CH}_3\text{OCH}_2\text{OCH}_3$

Ans: Dimethoxymethane

8) Diethyl ether does not react with sodium. Explain.

Ans: Diethyl ether does not contain any active hydrogen.

ASSERTION - REASONING QUESTIONS

A statement of assertion is followed by a statement of reason. Mark the correct choice from the options given below:

- (a) Both assertion and reason are true and reason is the correct explanation of assertion.
 - (b) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (c) Assertion is true but reason is false.
 - (d) Both assertion and reason are false.
-

1. **Assertion:** Methyl alcohol is a weaker acid than water.

Reason: Among the aliphatic monohydric alcohols, methyl alcohol is the strongest acid.

(Ans- b)

2. **Assertion:** o-Nitrophenol and p-Nitrophenol are separated by steam distillation.

Reason: o-Nitrophenol has intramolecular H-bonding while molecules of p-nitrophenol are linked by intermolecular H-bonding.

(Ans-a)

3. **Assertion:** Phenols do not react with phosphorus halides while alcohols do not react.

Reason: In phenols, the C-O bond has a partial double bond character due to resonance while it is not so in alcohols.

(Ans-a)

4. **Assertion:** Phenol is acidic in nature.

Reason: Phenate ion is less resonance stabilised than phenol.

(Ans- a)

One word answer

1. Name a compound that can be used as an anaesthetic in surgery.

(Ans - Ethrane)

2. In Williamson synthesis, which type of halide should not be used?

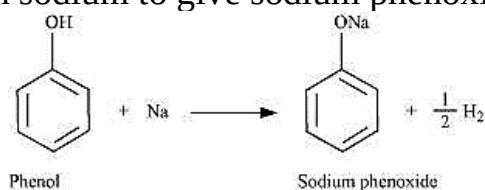
(Ans - Tertiary alkyl halide)

2 MARKS QUESTIONS

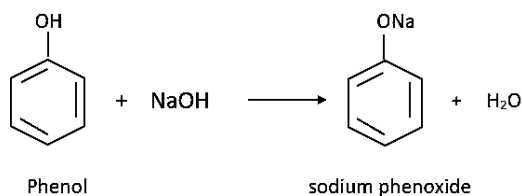
1) Give two reactions that show the acidic nature of phenol. Compare the acidity of phenol with that of ethanol.

A: The acidic nature of phenol can be presented by the following two reactions:

(i) Phenol reacts with sodium to give sodium phenoxide, liberating H_2 .



(ii) Phenol reacts with sodium hydroxide to give sodium phenoxide and water.



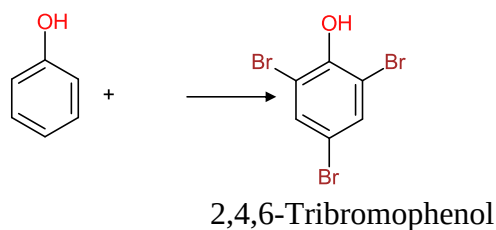
The acidity of phenol is more than that of ethanol. This is because after losing a proton, the phenoxide ion undergoes resonance and gets stabilised whereas the ethoxide ion does not.

2) How does phenol react with Br_2 in CS_2 and bromine water?

A: (i) When the reaction is carried out in solvents of low polarity such as $CHCl_3$ or CS_2 and at low temperature, monobromophenols are formed.

(ii) When phenol is treated with bromine water, 2,4,6-tribromophenol is formed as a white precipitate.





3) How do you account for the fact that unlike phenol, 2, 4-dinitrophenol and 2, 4, 6-trinitrophenol are soluble in aqueous solution of sodium carbonate?

A: 2,4-Dinitrophenol and 2, 4, 6-trinitrophenol are stronger acids than carbonic acid (H_2CO_3) due to the presence of the electron withdrawing-NO₂ group. Hence, they react with Na_2CO_3 to form their corresponding salts and dissolve in aq Na_2CO_3 solution.

4) Account for the following:

- (a) Boiling point of the $\text{C}_2\text{H}_5\text{OH}$ is more than that of $\text{C}_2\text{H}_5\text{Cl}$
- (b) The solubility of alcohols in water decreases with an increase in molecular mass.

A: (a) Because of hydrogen bonding.

(b) With an increase in molecular mass the non-polar alkyl group becomes more predominant.

5) Answer the following:

- a. What is the order of reactivity of 1, 2 and 3 alcohols with sodium metal?
- b. How will you account for the solubility of lower alcohols in water?

A: a. $1^\circ > 2^\circ > 3^\circ$

b. Here-OH group is predominant and the alcohol molecules can form hydrogen bonds with water molecules.

6) Give reasons:

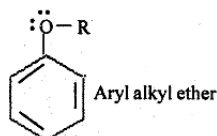
- i) Nitration of phenol gives ortho-and para-products only.
- ii) Why do alcohols have higher boiling points than haloalkanes of the same molecular mass?

A: i) -OH group increases the electron density more at ortho and para positions through its electron-releasing resonance effect.

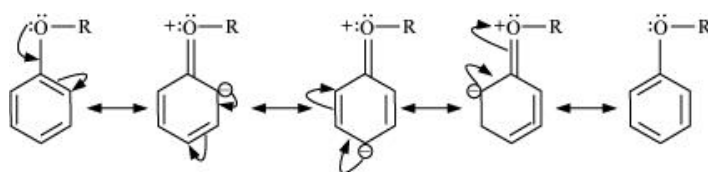
ii) Alcohols are capable of forming intermolecular H-bonds while alkyl halide doesn't.

7) Explain the fact that in aryl alkyl ethers

- (i) The alkoxy group activates the benzene ring towards electrophilic substitution and
- (ii) It directs the incoming substituents to ortho and para positions in the benzene ring.



A: (i) In arylalkylethers, due to the +R effect of the alkoxy group, the electron density in the benzene ring increases as shown in the following resonance structure.

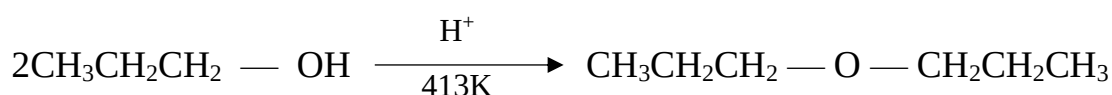


Thus, benzene is activated towards electrophilic substitution by the alkoxy group.

(ii) It can also be observed from the resonance structures that the electron density increases more at the ortho and para positions than at the meta position. As a result, the incoming substituents are directed to the ortho and para positions in the benzene.

8) How is 1-propoxypropane synthesized from propan-1-ol? Write the mechanism of this reaction.

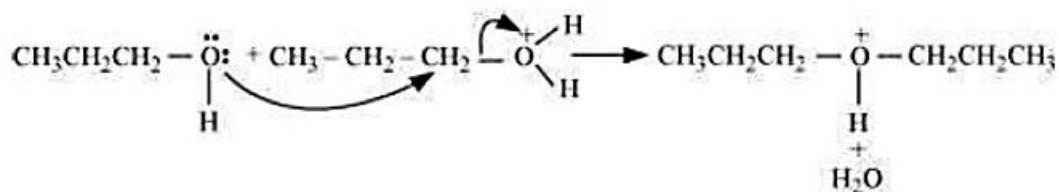
A: 1-propoxypropane can be synthesized from propan-1-ol by dehydration. Propan-1-ol undergoes dehydration in the presence of protic acids (such as H_2SO_4 , and H_3PO_4) to give 1-propoxypropane.



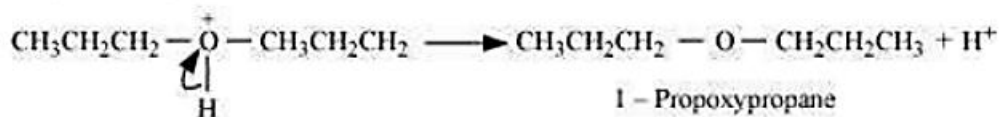
The mechanism of this reaction involves the following three steps.

Step 1: Protonation

Step 2: Nucleophilic attack



Step 3: Deprotonation



ASSIGNMENT

- Q1. How are primary, secondary and tertiary alcohols prepared from Grignard Reagents?
- Q2. Give the equations of oxidation of primary secondary and tertiary alcohols by Cu at 573K.
- Q3. Show how will you synthesize:
- 1-phenyl ethanol from a suitable alkene.
 - Cyclohexyl methanol using alkyl halide by $\text{S}_\text{N}2$ reaction.
 - Pentan-1-ol using a suitable alkyl halide?
- Q4. How are the following conversions carried out?
- Propene $\rightarrow$ Propan-1-ol
 - Benzyl chloride $\rightarrow$ Benzyl alcohol
 - Ethyl magnesium chloride $\rightarrow$ Propan-1-ol
- Q5. Name the reagents used in the following reactions:
- Oxidation of a primary alcohol to carboxylic acid.
 - Oxidation of a primary alcohol to aldehyde.
 - Bromination of phenol to 2,4,6-tribromophenol.

MULTIPLE CHOICE QUESTIONS (MCQs)

- 1) What is the IUPAC name of Vinyl alcohol?
- (a) Ethanol (b) Methanol

(c) Ethenol

(d) Methnol

ANS: (c)

2) Which of the following is more acidic than alcohol?

(a) Phenol

(c) Benzyl alcohol

(b) Cyclohexanol

(d) Ethenol

ANS: (a)

3) Which one of the following compounds is obtained by dehydrogenation of secondary alcohols?

(a) Ketone

(c) Carboxylic acid

(b) Aldehyde

(d) Amine

ANS: (a)

4) The reaction of carboxylic acid and alcohol catalysed by conc H_2SO_4 is called?

(a) Dehydration

(c) Esterification

(b) Saponification

(d) Neutralization

ANS: (c)

5) Which of the following alcohols is most soluble in water

(a) Propanol

(c) Pentanol

(b) Hexanol

(d) Butanol

ANS:(a)

6) On heating aqueous solution of benzene diazonium chloride, which of the following is formed

(a) Benzene

(c) Phenol

(b) Chlorobenzene

(d) aniline

ANS:(c)

7) Catalytic dehydrogenation of a primary alcohol gives a

(a) Ketone

(c) Sec. alcohol

(b) Aldehyde

(d) Ester

ANS:(b)

8) Ethyl alcohol obtained by fermentation of starch is called wash and what is its purity?

- (a) 15% (c) 99.9%
(b) 99% (d) 95%

ANS:(a)

9) Which chemical is used to distinguish between phenol and benzyl alcohol?

- (a) NaHCO_3 (c) Iodoform test
(b) FeCl_3 (d) none of the above

ANS:(b)

10) Which is most acidic

- (a) Phenol (c) Cresol
(b) 4-nitrophenol (d) 2-nitrophenol

ANS:(d)

SHORT ANSWERED QUESTIONS [1-MARK]

1) Write the IUPAC names of $\text{CH}_2-(\text{CH})\text{CH}_2\text{OH}$

2) Which of the following has higher pK_a value: Nitrophenol OR phenol

3) Arrange the following compounds in order of increasing boiling points.

- (i) Bromoethane (ii) Ethanol (iii) Methoxymethane

4) Write the structure of 2-Ethoxy-3-methyl pentane

5) Write the product formed when Ethoxy benzene reacts with HI

6) Which is more acidic: ortho-nitrophenol or ortho-methoxyphenol?

7) Name the chemical test used to distinguish between Phenol and Ethanoic acid.

8) Write the name of the product of the reaction of Bromine in CS_0 with phenol

9) What is the condition of the compound to undergo a Iodoform test?

10) Write the structure of cumene.

ANSWERS

1) Prop-2-en-1-ol

2) Phenol

3) 3. (i)<(iii)<(ii)

4) 4. $\text{CH}_3\text{-CH-CH}[\text{CH}_3]\text{-CH}[\text{OC}_2\text{H}_5]\text{-CH}_3$

5) 5 Phenol + Iodoethane

6) ortho-nitrophenol (Due to the electron-withdrawing group)

7) FeCl_3 test or sodium bicarbonate test

8) o-bromophenol and p-bromophenol

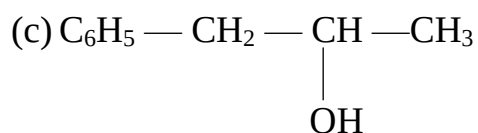
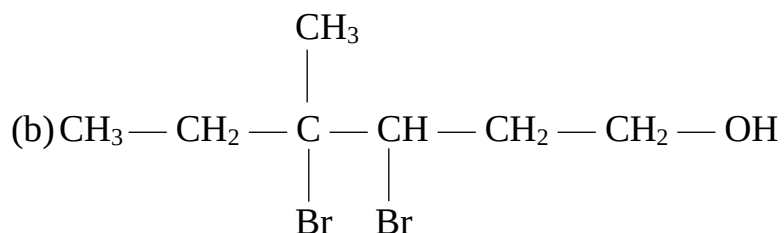
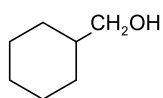
9) presence of terminal methyl group with alcoholic/carbonyl group

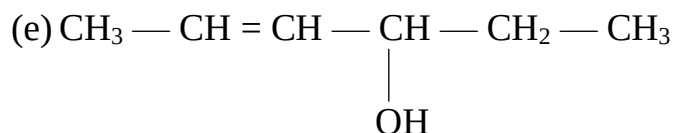
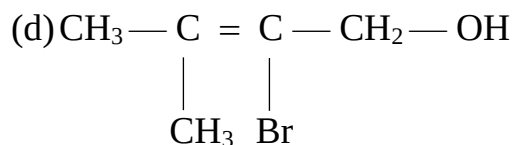
10) $\text{CH}_3\text{-CH}[\text{C}_6\text{H}_5]\text{-CH}_3$ [Isopropyl benzene]

BOARD QUESTIONS

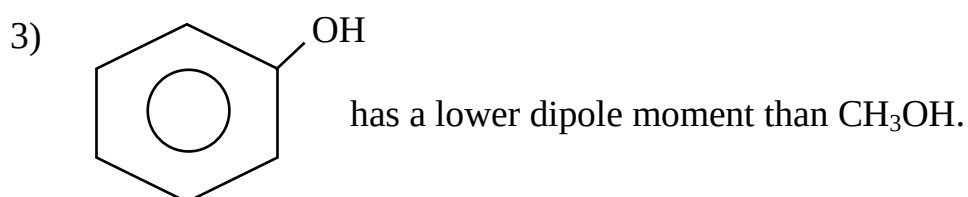
1) Write the IUPAC name of the following:

(a)

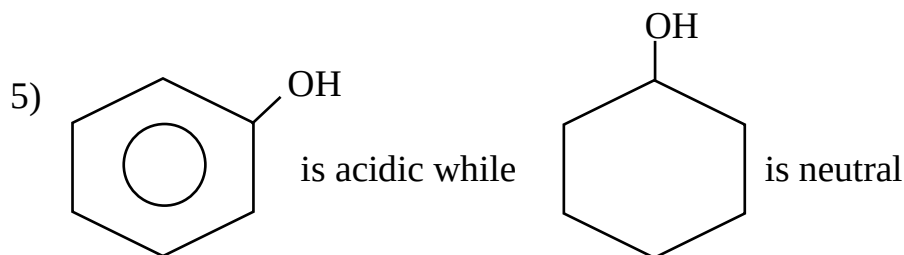




2) Phenols do not undergo S_N to the $-\text{OH}$ group like alcohols?



4) Phenol doesn't undergo protonation like that of ROH . Why?



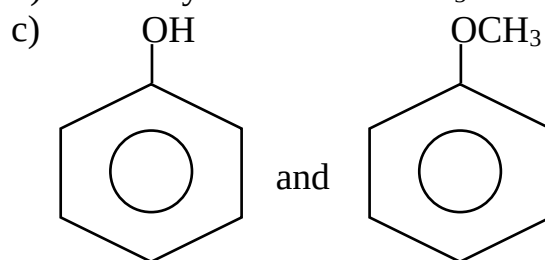
6) The Boiling Point of ROH is more than that of ethers.

7) Dehydration of ROH to form alkene is carried out with conc. H_2SO_4 and not with conc. HCl .

8) Distinguish between:

a) Propanol-1 and Propanol-2

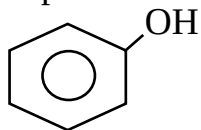
b) Tert-butyl alcohol and CH_3OH



d) Phenol and Benzoic acid

9) Convert the following:

- a) 2-Bromopropane to acetone
- b) Acetone to tert-butyl alcohol
- c) But-2-ene to ethanol
- d) Ethanol to CH_3COOH
- e) Phenol to salicylic acid & Salicylaldehyde
- f) Propan-2-ol to Propan-1-ol
- g) Ethanol to Propan-2-ol
- h) Propene to Propan-1-ol
- i) Phenol to Benzoquinone
- j) Propene to Propan-2-ol
- k) Anisole to



l) Benzyl alcohol to Benzoic acid

10) Write the structure of 2-Methyl-2-ethoxy-pentane.

11) Convert Propanone to 2-methylpropan-2-ol

12) Write the chemical equation for the preparation of phenol from C_6H_6 using oleum and NaOH .

13) Give Reason:

- a) Preparation of ethers by acid dehydration of sec or tert alcohols is not a suitable method.
- b) o-nitrophenol is more acidic than methoxyphenol

14) Draw the structure and names of the products formed if the following alcohols are oxidised. Assume that an excess of OH is used.

- (i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- (ii) 2-Butanol
- (iii) 2-Methyl-1-propanol

15) List two strong oxidising and reducing agents and also two weak oxidising and reducing agents.

16) Show how will you synthesize:

- (a) 1-phenylethanol from a suitable alkene
- (b) Cyclohexylmethanol using an RX by an $\text{S}_{\text{N}}2$ reaction

(c) Pentane-1-ol using a suitable RX

17) How are the following conversions carried out?

(i) Ethylmagnesium chloride to Propan-1-ol

(ii) Methylmagnesium bromide to 2-methyl-propan-2-ol

18) $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3$ on reaction with HI gives $(\text{CH}_3)_3\text{C}-\text{I}$ and CH_3-OH as the main products and not $(\text{CH}_3)_3\text{C}-\text{OH}$ and CH_3I .

19) Convert phenol to m-bromophenol

20) Give Reason: Phenol does not react with NaHCO_3 whereas RCOOH react.

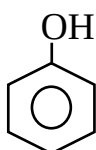
21) Which is a stronger acid Phenol or Cresol?

22) Give Reason:

a) $(\text{CH}_3)_3\text{C}-\text{Br}$ on reaction with NaOCH_3 gives an alkene instead of an ether.

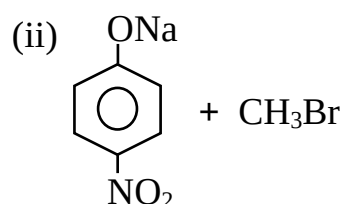
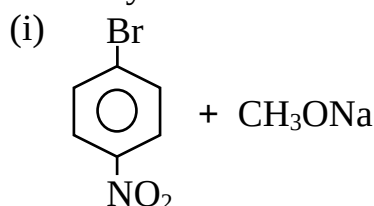
b) Relative ease of dehydration of ROH is $3^\circ > 2^\circ > 1^\circ$.

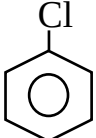
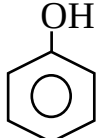
c) o-nitrophenol is more acidic than o-methoxyphenol.

23) Convert  to anisole

24) Relative ease of dehydration of alcohols is $3^\circ > 2^\circ > 1^\circ >$

25) Which of the following is an appropriate set of reactants for the preparation of 1-methoxy-4-nitrobenzene & why?



26) Convert:  to 

27) Convert Benzylalcohol to Benzoic acid

28) While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which is steam volatile, giving reason.

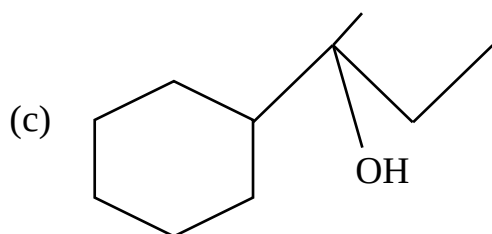
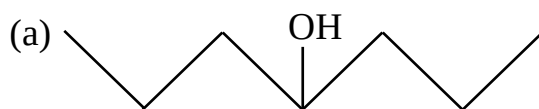
29) Write the names of reagents and equations for the preparation of the following ethers by willaimsons synthesis.

(a) 2-Methoxy-2-methylpropane

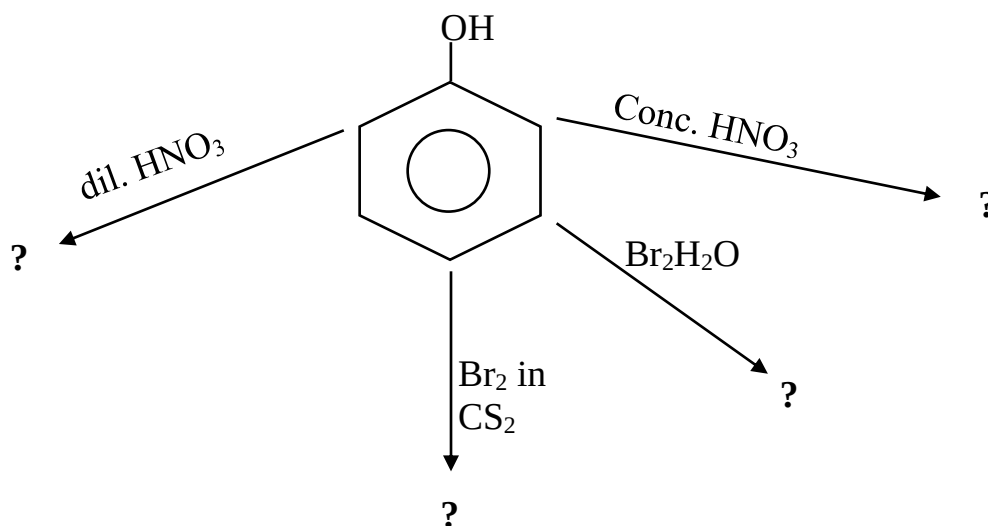
(b) 1-Methoxyethane

30) What is the limitation of willaimson synthesis for the preparation of certain types of ethers?

31) How will you synthesise the following alcohols from the appropriate alkene?



32) Complete the following:



- 33) Write the structure of the major products obtained by
- (a) Mononitration of 3-Methylphenol
 - (b) Dinitration of 3-Methylphenol
 - (c) Mononitration of Phenylmethanoate
- 34) Write the equation for the reaction of HI with
- a) 1-Propoxypropane
 - b) Methoxybenzene
 - c) Benzylethylether
- 35) How is 1-Propoxypropane synthesised from propan-1-ol. Give the mechanism.
- 36) Synthesise the following by williamsons synthesis
- (a) 1-Propoxypropane
 - (b) Ethoxybenzene

_____ X _____ X _____ X _____